AA/ME 507 Winter 2026
Homework 4
February 24, 2026
Due Date: March 4, 2026 (11:59 PM)
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Problem 1. Boussinesq approximation and dimensional analysis: designing a lava lamp
A lava lamp contains a *wax* phase dispersed as approximately spherical blobs of radius R moving through an *ambient liquid* of density ρ_ℓ and dynamic viscosity μ. Gravity acts downward with magnitude g. The wax–liquid interfacial tension is σ. The bottom of the lamp is warm at temperature T_h and the top is cool at temperature T_c with $\Delta T \equiv T_h - T_c > 0$.
Assume the Reynolds number based on the blob is small (creeping flow), and that density variations are small enough for a Boussinesq approximation. Model the wax density as

$$\rho_w(T) \approx \rho_{w,c}\big(1 - \beta(T - T_c)\big),$$

so that a characteristic buoyancy-producing density difference scales as

$$\Delta\rho \sim \rho_{w,c}\,\beta\,\Delta T_{\text{drive}},$$

where ΔT_{drive} is the relevant temperature mismatch between the blob and its surroundings. The wax has thermal diffusivity α (units L^2/T). Ignore numerical prefactors throughout (use scaling balances only).

Iteration 1 (two-temperature model) Assume the ambient liquid is everywhere at T_c, while a rising wax blob is initially at T_h.

1. Estimate the blob radius R by balancing buoyancy against surface tension.
2. Estimate the thermal equilibration time scale t_{th} for the blob to cool substantially toward T_c.
3. Estimate the terminal rise speed U using Stokes drag.
4. Estimate the lamp height H such that the blob *just loses buoyancy* near the top, by imposing $H/U \sim t_{\text{th}}$.

Iteration 2 (stratified ambient) Now assume the ambient temperature varies linearly with height:

$$T(z) = T_h + (T_c - T_h)\frac{z}{H} = T_h - \Delta T\,\frac{z}{H}, \qquad 0 \le z \le H,$$

where $z = 0$ at the bottom and $z = H$ at the top. Define the local temperature mismatch

$$\Delta(t) \equiv T_b(t) - T(z(t)),$$

where $T_b(t)$ is the (approximately uniform) blob temperature and $z(t)$ its height.

1. Using a relaxation model for $T_b(t)$, estimate an *effective* driving contrast ΔT_{eff} that sets buoyancy in the stratified case.
2. Using ΔT_{eff} in place of ΔT where appropriate, restate the scalings for R, t_{th}, U, and H.

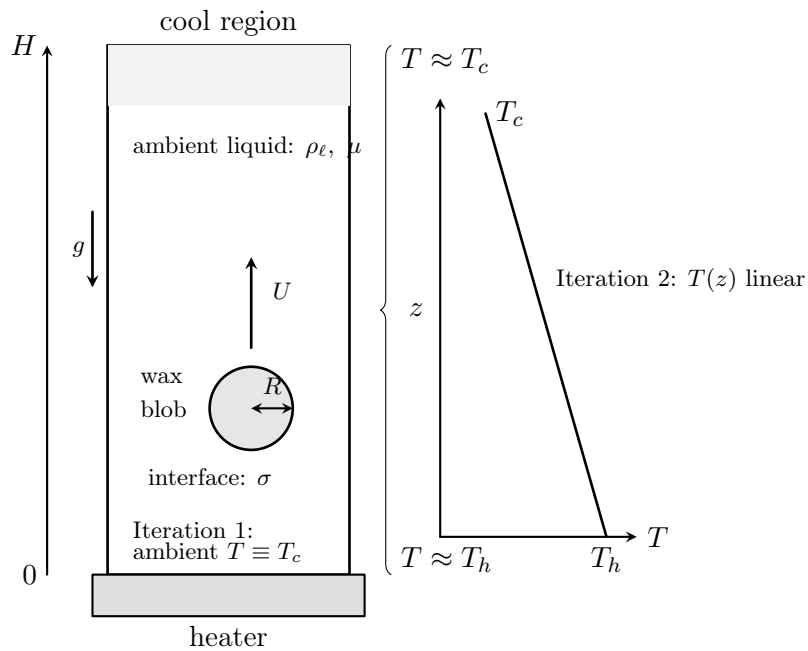

Figure 1: Schematic of the lava-lamp scaling model.

Problem 2. Pressure, density, incompressibility, and the speed of sound Let's derive an evolution equation for the pressure in a fluid starting from the mass conservation equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

where ρ is the density, $\mathbf{v}$ is the velocity field, and t is time.

1. Expand the divergence term in the previous using the product rule, and use the definition of the speed of sound c in a fluid,
$$c^2 = \left(\frac{\partial p}{\partial \rho}\right)_s,$$
where p is pressure and the derivative is taken at constant entropy s, to replace the $\nabla\rho$ term in that equation with a term proportional to ∇p.

2. Group the pressure terms in the resulting equation to find an equation in terms of Dp/Dt and $\nabla \cdot \mathbf{v}$.

3. For incompressible flow, $c = \infty$ and $\nabla \cdot \mathbf{v} = 0$. In these flows, what does mass conservation tell us about the dynamics of p?

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Problem 3. Vortex stretching in an axisymmetric stagnation flow. Consider an incompressible axisymmetric straining flow

$$u_r = -\frac{\alpha}{2} r, \qquad u_z = \alpha z,$$

with no swirl in the base flow. Let the azimuthal vorticity be $\omega_\theta(r, z, t)$.

1. Derive the evolution equation for ω_θ, including viscosity.

2. Solve the inviscid equation along pathlines to show exponential amplification/decay associated with vortex stretching.

Problem 4. Biot-Savart Law: Velocity jump across a vortex sheet. A 2D vortex sheet lies along a smooth curve $\boldsymbol{x} = \boldsymbol{x}(s)$ (arclength s) with sheet strength $\gamma(s)$, such that $\omega(\boldsymbol{x}) = \int \gamma(s)\,\delta(\boldsymbol{x} - \boldsymbol{x}(s))\,\mathrm{d}s$ (see figure 2).

1. Starting from the 2D Biot–Savart law, find the line integral of $\gamma(s)$ that provides the velocity induced by the vortex sheet.

2. Assume that γ can be considered as locally constant to prove that the tangential velocity has a jump of magnitude γ across the sheet.

3. Specialize to a flat sheet on $y = 0$ and express the induced normal and tangential velocity components.

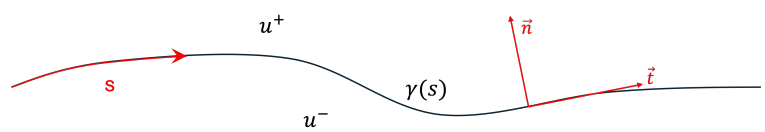

Figure 2: Vortex sheet.

Problem 5. Rankine vortex: vorticity and pressure deficit A common idealized model for compact swirling flows (e.g. dust devils, tornado cores, hurricane eyewall structure) is the *Rankine vortex*: solid-body rotation inside a core of radius a, and a potential vortex outside. Assume steady, incompressible, axisymmetric flow in cylindrical coordinates (r, θ) with no radial velocity ($u_r = 0$) and tangential velocity

$$u_\theta(r) = \begin{cases} \Omega r, & 0 \le r \le a, \\ \dfrac{\Gamma}{2\pi r}, & r \ge a, \end{cases}$$

where Ω and Γ are constants and ρ is the (constant) density.

1. Verify that this velocity field is incompressible.

2. Compute the out-of-plane vorticity

$$\omega_z(r) = \frac{1}{r}\frac{d}{dr}\big(r u_\theta(r)\big)$$

for $0 \le r \le a$ and for $r \ge a$. Sketch $\omega_z(r)$.

3. Impose continuity of u_θ at $r = a$ to relate Γ and Ω. Give a brief physical interpretation of this matching.

4. For a purely swirling steady inviscid flow with $u_r = 0$, the radial momentum balance is

$$\frac{dp}{dr} = \rho\,\frac{u_\theta^2}{r}.$$

Integrate this equation to obtain $p(r) - p(\infty)$ for $r \ge a$, and then determine the center pressure drop $p(0) - p(\infty)$ (assume p remains finite at $r = 0$). Express your final answer in terms of ρ, a, and either Ω or Γ.

5. Determine the radius where u_θ attains its maximum value, and express $u_{\theta,\max}$ and the center pressure drop in terms of $u_{\theta,\max}$, ρ, and a.